CPSC 100

Computational Thinking

Data Representation

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Agenda

- Course Admin
  - (Dr. Moosvi will post any important reminders on Ed Discussion)

- Intro to Data Representation

Learning Goals

After this today's lecture, you should be able to:

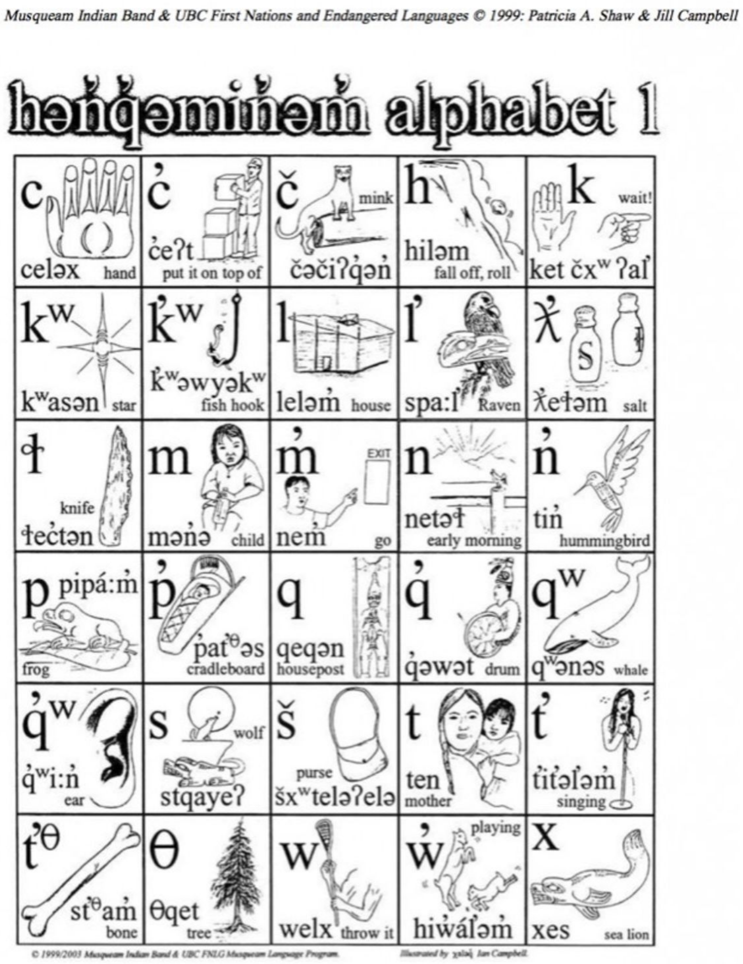
- Recognize binary’s role in data representation
- Understand that binary (base-2) is the fundamental numbering system used in computing.
  - Explain why computers use binary
- Convert given decimal numbers (e.g., 13) into binary.
- Perform a binary “magic trick” to identify numbers using logical reasoning
- *Bonus: Count up to 32 using 1 hand*

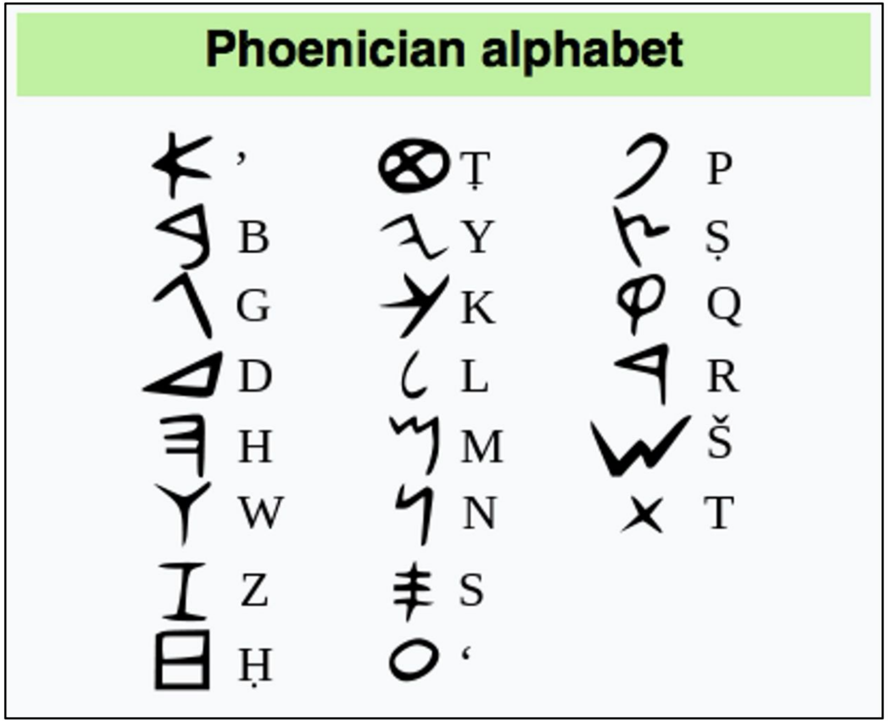
Data Representation Before Computers

Digital Data Before Computers

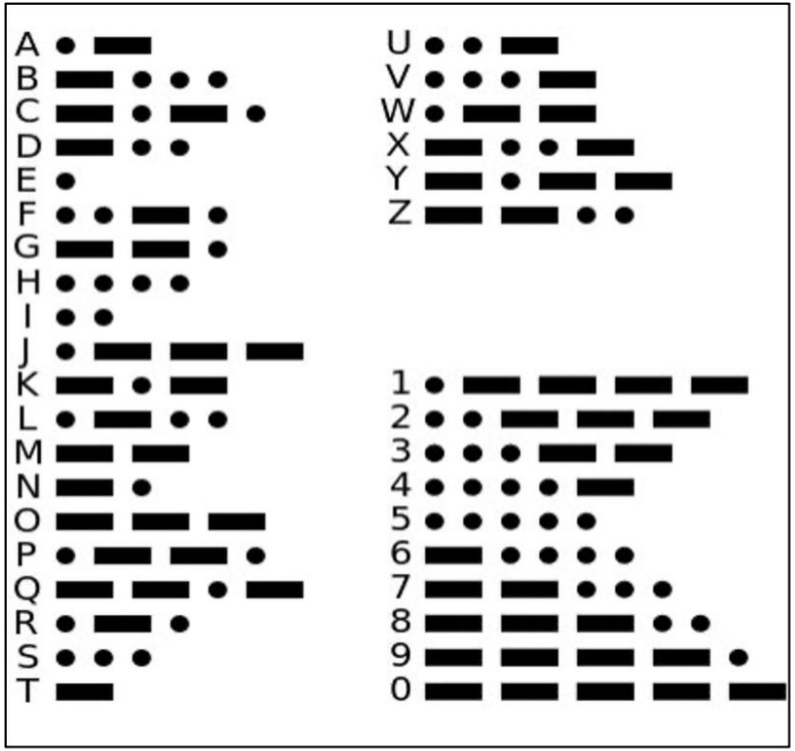
Creative computational thinkers have invented digital data representation systems for millennia, adapting to many different technologies:

Image source: Wikipedia

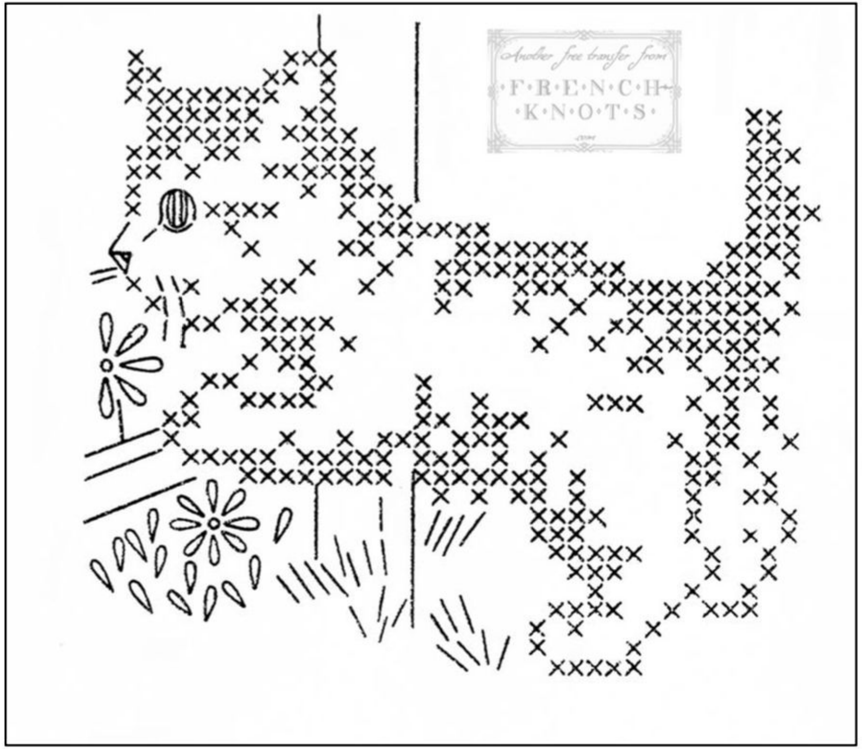




Source: Musqueam Teaching and Learning



Morse Code



Cross Stitch

Image source: Wikipedia

Participation Question

How do you count from 1 to 10 on your hands?

Any volunteers to come show it on the document camera?

Data Representation **in nature!**

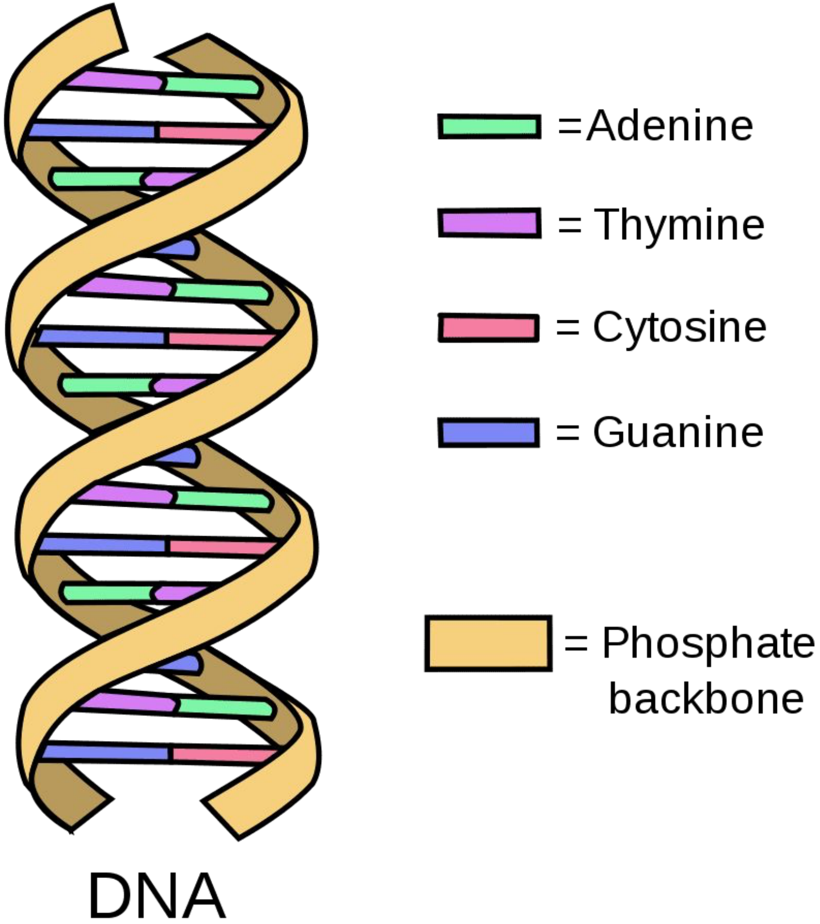


Image source: ashg.org

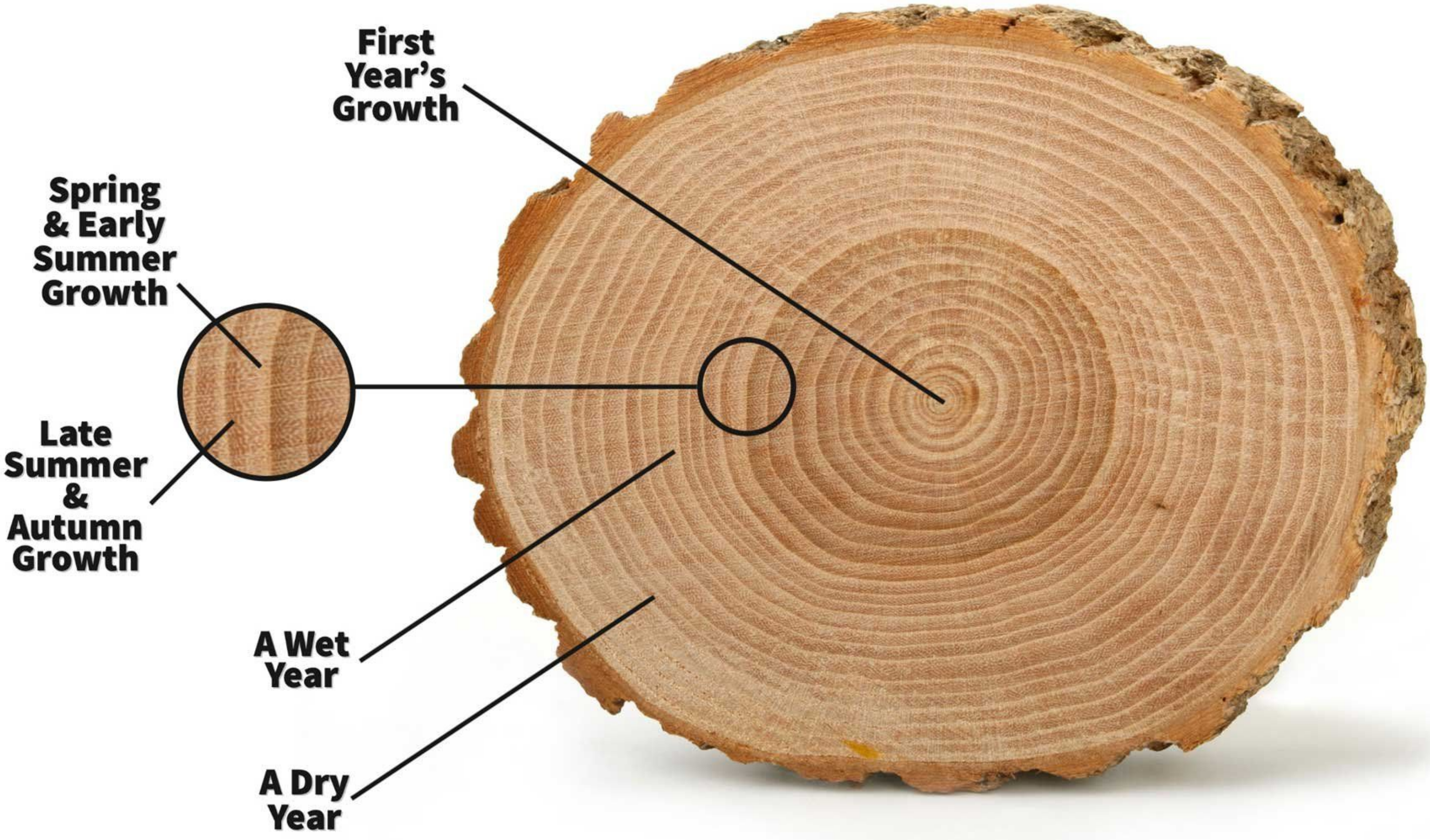


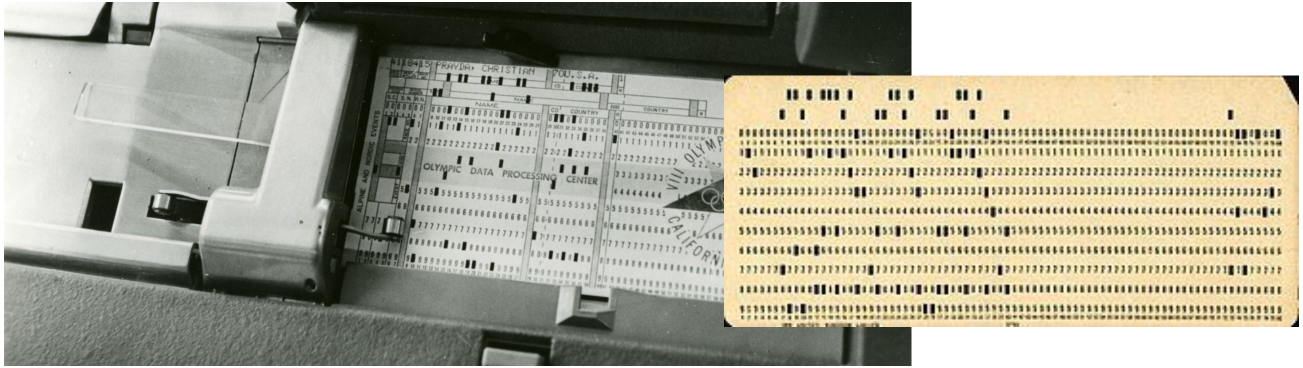
Image source: silvotherapy.co.uk

Data Representation with computers

Digital Data with Computers

- Computing technologies represent numbers, text, images, video, and sound using just two symbols: **0's and 1's**

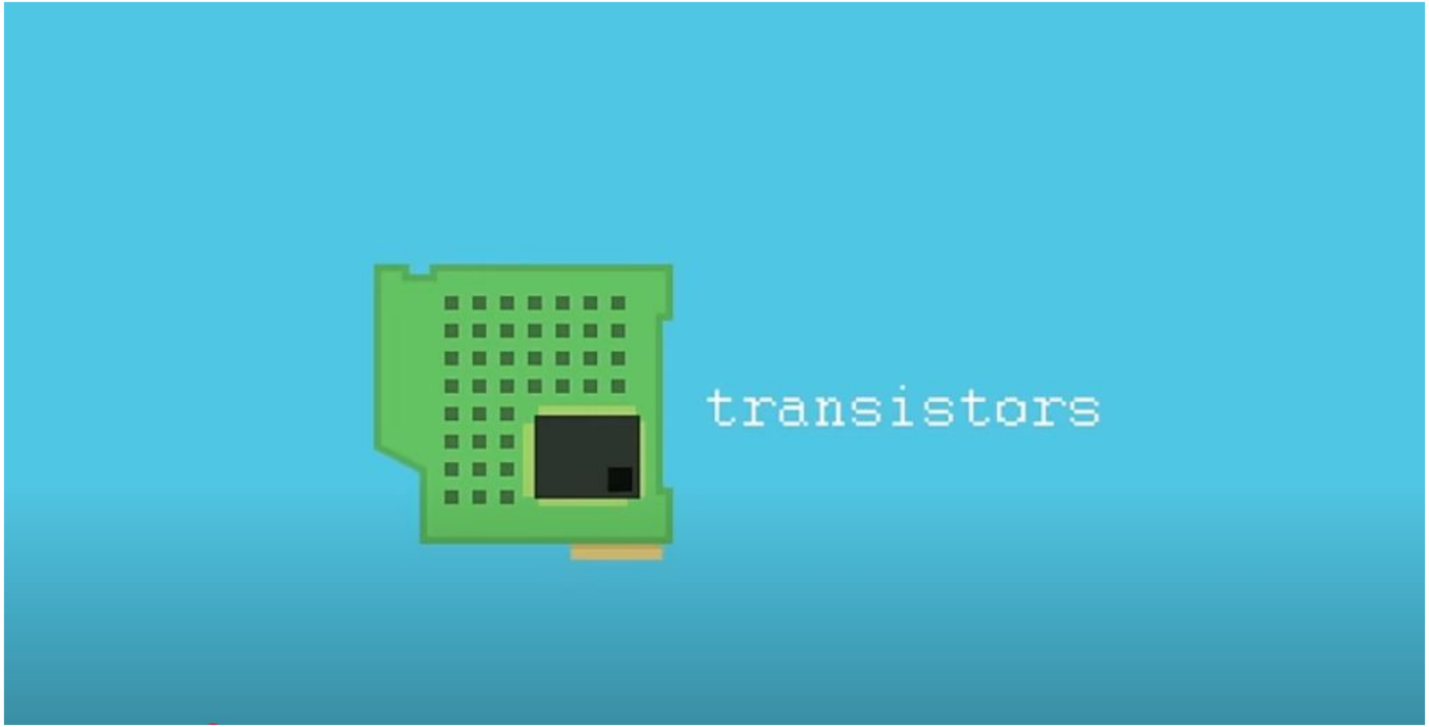
- 0's and 1's are embodied as "high" (on) or "low" (off) signals on various electronic and optical storage media (e.g., punch cards, vacuum tubes, transistors, DVDs, etc.)



high	low
on	off
true	false
yes	no
+	-
1	0

Image source: Wikipedia

Required Video: Why binary?



Numbers in a Digital Era

- [Base-10] Decimal Numbers
    - E.g., 1, 2, 3, etc.
- [Base-2] Binary Numbers
    - E.g., 11001, 1001, 1111, etc.
- [Base-3] Ternary Numbers
    - E.g., 12_3, 110_3, 20_3, etc.
- [Base-16] Hexadecimal Numbers
    - E.g., FF5733, AB4151, etc.

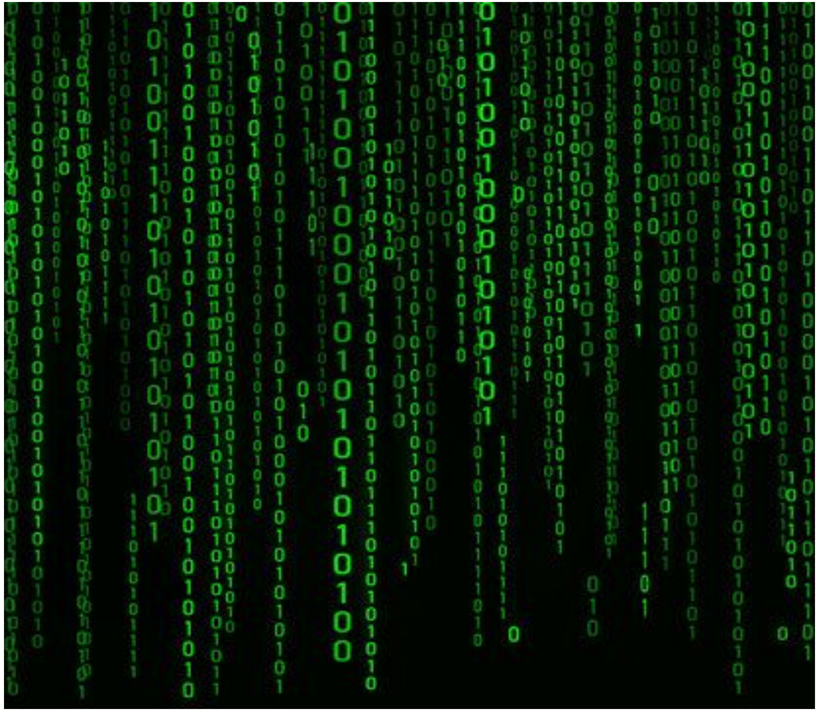


Image source: Adobe

Decimal Numbers

Decimal Numbers

Decimal numbers are base 10.

("decem" is Latin for 10).

In decimal, every column is worth 10 times as much as the previous one

2 4 8

Each digit can be any one of these 10 values: {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}

1s place (10^0)

10s place (10^1)

100s place 100 (10^2)

Counting in Decimal

When counting in decimal:

- Add one to the right most digit
- If that digit has the highest value (9), then add one to the digit in the next column to the left and reset the column you're on to zero.
- If you run out of digits in the column to the left, you repeat the process

- 0
- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8
- 9

...

99

100

...

10999

11000

Counting in Decimal Example 1

1	0	0	+1	→	1	0	1

Add 1 to the
rightmost digit

Counting in Decimal Example 2

1 0 9 +1 → 1 1 0

Add 1 to the rightmost digit. 9 is the highest representation for a number, so we have to change 9 to 0 and increment the number to its left by 1

Binary Numbers

Bits

- A **bit** is short for a **binary digit**.
  - It is one 0 or 1. E.g., one "high" segment on a DVD or an "on" transistor.
- When a computer sees 1000000 (e.g. one "on" part followed by six "off" parts), it reads this as **binary** and *processes* this as **decimal 64**.
  - 64 and 1000000 are two representations of the same number
- When reading out binary numbers, **we say the digits in turn**
  - For “010” we say "zero one zero" or just "one zero“ (We don't say "ten").
  - Sometimes we write this as **0b010** to make it clear it is a **binary** number.
  - You may also see a subscript 2 after the number (e.g., 010_2).

Binary Numbers

Binary numbers are base 2.
("binarius" is Latin for having two parts).
In binary, every column is worth 2 times as much as the previous one

1 1 0 0 1 ← Binary

1s place (2^0)
2s place (2^1)
4s place (2^2)
8s place (2^3)
16s place (2^4)

Convert Binary to Decimal

0b 1 1 0 0 1

1s place multiply by 1	(2^0):	1
2s place multiply by 2	(2^1):	0
4s place multiply by 4	(2^2):	0
8s place multiply by 8	(2^3):	8
16s place multiply by 16	(2^4):	+ 16
Total in decimal (add them up):		25

Algorithm to Convert Decimal to Binary

- Start with a decimal number *n* (0 to 255)
  - Since 255 is the maximum, we need 8 bits.
- Check the highest power of 2 that fits into *n*:
  - If $n < 128$, set the 1st (leftmost) bit to 0; otherwise, set it to 1 and subtract 128 from *n* ($n=n-128$)
  - If $n < 64$, set the 2nd bit to 0; otherwise, set it to 1 and subtract 64 from *n*. ($n=n-64$)
  - If $n < 32$, set the 3rd bit to 0; otherwise, set it to 1 and subtract 32 from *n*. ($n=n-32$)
  - Continue this process for 16, 8, 4, 2, and finally 1.
- By the end, *n* will be either 0 or 1, which is the last (8th) bit.

Algorithm to Convert 217 to Binary

- n = 217
- Check the highest power of 2 that fits into *n*:
  - ❌ **217** < 128 (2^7), 1st Bit = 1
    - *n = 217 - 128* ⇒ 89
  - ❌ **89** < 64 (2^6), 2nd bit = 1
    - *n = 89 - 64 ⇒ 25*
  - ✅ **25** < 32 (2^5), 3rd bit = 0
    - *No subtraction required*
  - ❌ **25** < 16 (2^4), 4nd bit = 1
    - *n = 25 - 16 ⇒* 9
  - ❌ **9** < 8 (2^3), 5th Bit = 1
    - *n = 9 - 8 ⇒* 1
  - ✅ **1** < 4 (2^2), 6th bit = 0
    - *No subtraction required*
  - ✅ **1** < 2 (2^1), 7th bit = 0
    - *No subtraction required*
  - **Final bit = n = 1 (2^0)→ 8th bit = 1**
- **Final Answer: 11011001**

Binary Numbers

- Binary numbers only use two digits: 0 and 1, and their place values are based on powers of 2.

Number 9 in Binary:

256	128	64	32	16	8	4	2	1
0	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**1**

9 = 000001001

Counting in Binary

When counting in binary:

- Add one to the right most digit
- If that digit has the highest value (1), then add one to the digit in the next column to the left and reset the column you're on to zero.
- If you run out of digits in the column to the left, you repeat the process

- 0
- 1
- 10
- 11
- 100
- 101
- …
- 101101
- 101110
- 110001

Counting in Binary Example 1

1 0 0 +1 → 1 0 1

Add 1 to the rightmost digit

Clicker: Addition in Binary

iClicker

$\underline{\;1\;}\ \underline{\;0\;}\ \underline{\;1\;}\ +1 \rightarrow \underline{\;?\;}\ \underline{\;?\;}\ \underline{\;?\;}$

What number should go on the right side?
(note: both #s are in binary)

A) 100
B) 101
C) 110
D) 111

iClicker

Q: The 8-bit binary representation of 57 is 00111001. What is the 8-bit binary representation of 58?

A. 01011110

B. 00111111

C. 00111010

D. 0011100010001

Wrap up

Take-Home Practice

Convert these numbers to binary

- 16
- 27
- 30
- 58
- 98
- 78
- 100
- 9
- 34